BARUN RANJAN CHHETRI CHAMLAGAIN

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PROFESSIONAL SUMMARY

Computer Science student with hands-on experience in data acquisition, data ingestion, and data reconciliation, building full-stack applications and data pipelines with Java, Spring Boot, Python, SQL, React, Next.js, PostgreSQL, and PGVector. Skilled in document parsing, text normalization, embedding-based entity resolution, semantic search, Retrieval-Augmented Generation (RAG), and REST API integration. Experienced turning messy, incomplete data into structured, reliable outputs, with unit and integration testing and Docker deployment. Seeking a Data Acquisition Engineer or Software Engineering / AI-ML internship starting May 2027.

TECHNICAL SKILLS

Languages: Java, Python, SQL, JavaScript, TypeScript
Frameworks and Libraries: Spring Boot, Spring AI, React, Next.js, Tailwind CSS, Prisma ORM, Apache PDFBox, Apache POI
Data and AI: PostgreSQL, PGVector, Vector Database, Vector Search, Semantic Search, Embeddings, Retrieval-Augmented Generation (RAG), Large Language Models (LLM), Generative AI, Machine Learning, Deep Learning, ETL, Data Pipelines, Data Ingestion, Data Acquisition, Data Normalization, Entity Resolution
Tools and Practices: REST API, OpenAPI / Swagger, Docker, Docker Compose, Git, GitHub, Amazon Web Services (AWS), Flyway, JUnit, Vitest, Unit Testing, Integration Testing, JWT Authentication, BCrypt

PROJECTS

AI Resume Intelligence Platform | Java 21, Spring Boot, Spring AI, PostgreSQL, PGVector, React, Docker
Personal Project | *Spring 2026 - Present* | *https://resume-analyzer.barunchamlagain.me/*

- Built a full-stack platform for data acquisition and ingestion of resumes (PDF, DOCX, TXT) and job postings, reconciling them into a searchable PostgreSQL/PGVector knowledge base using Apache PDFBox and Apache POI for text extraction.
- Designed a 4-tier entity resolution and skill-matching engine (exact, normalized, alias, and cosine-similarity semantic search on 1536-dimensional embeddings) backed by a canonical taxonomy of 100+ skills and an alias table, so variants like “AWS” and “Amazon Web Services” resolve to one entity.
- Implemented skill-gap analysis that separates critical from preferred gaps and surfaces adjacent evidence, plus grounded Retrieval-Augmented Generation (RAG) recommendations constrained to verified resume content to prevent fabricated experience.
- Managed schema with Flyway migrations and HNSW vector indexes; documented the REST API with OpenAPI/Swagger; wrote unit and MockMvc integration tests; containerized the stack with Docker Compose.

180-Day Interview Prep Tracker | Next.js 16, React 19, TypeScript, Prisma, PostgreSQL, Docker
Personal Project | *October 2026 - Present* | *https://interview-crack.barunchamlagain.me/*

- Built a full-stack, deployed web app that turns a 180-day study handbook into 900 tasks and 965 spaced-repetition review intervals, parsed from a PDF and seeded into PostgreSQL via Prisma.
- Implemented a timestamp-based study timer that survives refreshes, partial task carryover with undo and duplicate prevention (wrapped in database transactions), and daily/weekly/monthly reports emailed via Resend or SMTP.
- Secured the app with BCrypt password hashing, signed JWTs in HTTP-only cookies, login rate limiting, single-use password-reset tokens, and per-user data isolation; verified with a 26-scenario Vitest suite.

EDUCATION

Bachelor of Science in Computer Science | University of North Texas, Denton, TX | Expected May 2028
GPA: 3.850

HONORS AND CERTIFICATIONS

- **Dean's List:** Tarrant County College (Spring, Summer, Fall 2025; Spring 2026); University of South Dakota (Fall 2024)
- **Certifications:** Google: Introduction to Large Language Models (May 2026); LinkedIn: Introduction to Artificial Intelligence (Sep 2026); LinkedIn: Cybersecurity Foundations (Sep 2026)

LEADERSHIP AND COMMUNITY INVOLVEMENT

- Member, Mentors, Mindset, & Money: Intercultural Network, TCCD | Spring 2026